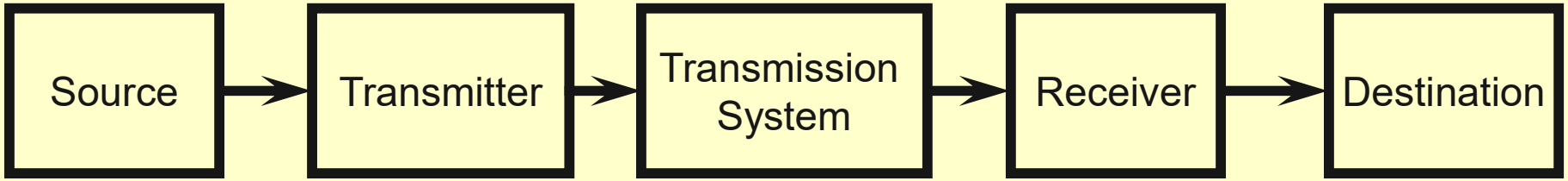
Protocol Hierarchies

- Computer networks are generally comprised of numerous pieces of hardware and software.
- Recall the previous attempt to ‘model’ how these components interconnect and interact with each other.



The Five-component Communications Model.

Protocol Hierarchies

- Whilst this horizontal model is useful when looking at hardware components, the reality is it does not explain everything.
- From previous discussions on: ***Data Link Control***, ***LANs technologies***, ***Internetworking*** etc. much of the functionality required to provide communications between host machines is not necessarily provided on a single hardware component:
  - For instance, the transmission/reception of ***IP Datagrams*** and ***MAC frames*** requires ultimately requires the ***encoding/decoding*** marks & spaces on a variety of transmission media using a variety of access techniques.
  - This mixture of functionality is provided on, possibly multiple ***NICs*** (one per LAN technology) as well as software elsewhere on the host.
  - This is not easily discernible from the horizontal Five-component Communications Model.

Protocol Hierarchies

- A better model is required to explain these concepts and techniques.
- To simplify network design most networking technologies are organized as ***layers*** of protocol software:
  - The purpose of each layer is to provide <u>one</u> component of the overall solution to the “Communications Problem” such as “frame reception/transmission” etc.
  - These layers of software can exist on separate hardware components and/or hosts or is some instances multiple layers can exist on a single hardware platform and/or hosts.

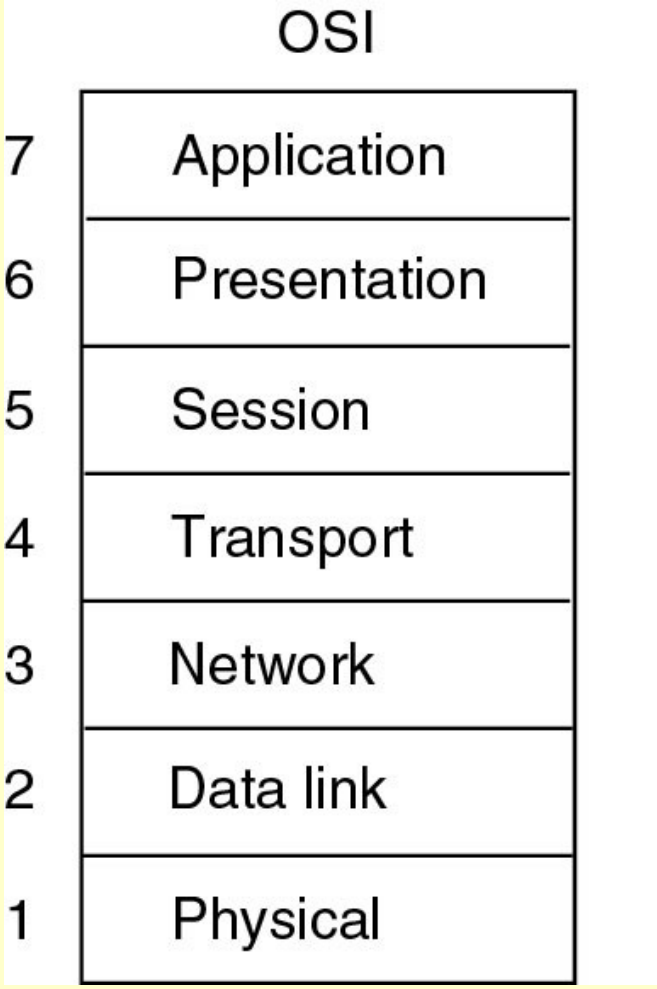
Protocol Hierarchies

- This concept is not dissimilar to the approach used for software development:
  - Objects or libraries are used to perform specific operations.
  - Importantly the object/library function keeps details of its internal state and algorithms hidden from the main program.
- The same approach is used for network protocol software:
  - Each layer of software provides one part of the solution such as: frame transmission/reception or, transmitting one/zeros onto a wire/wireless transmission medium etc.

Protocol Hierarchies

- With networking software, the layers of protocol software are organised into a ***Stack***:
  - Each layer sits between other layers as shown in the next slide.
  - Each layer is said to offer ***services*** to higher layers and to use the services offered by the lower layers.

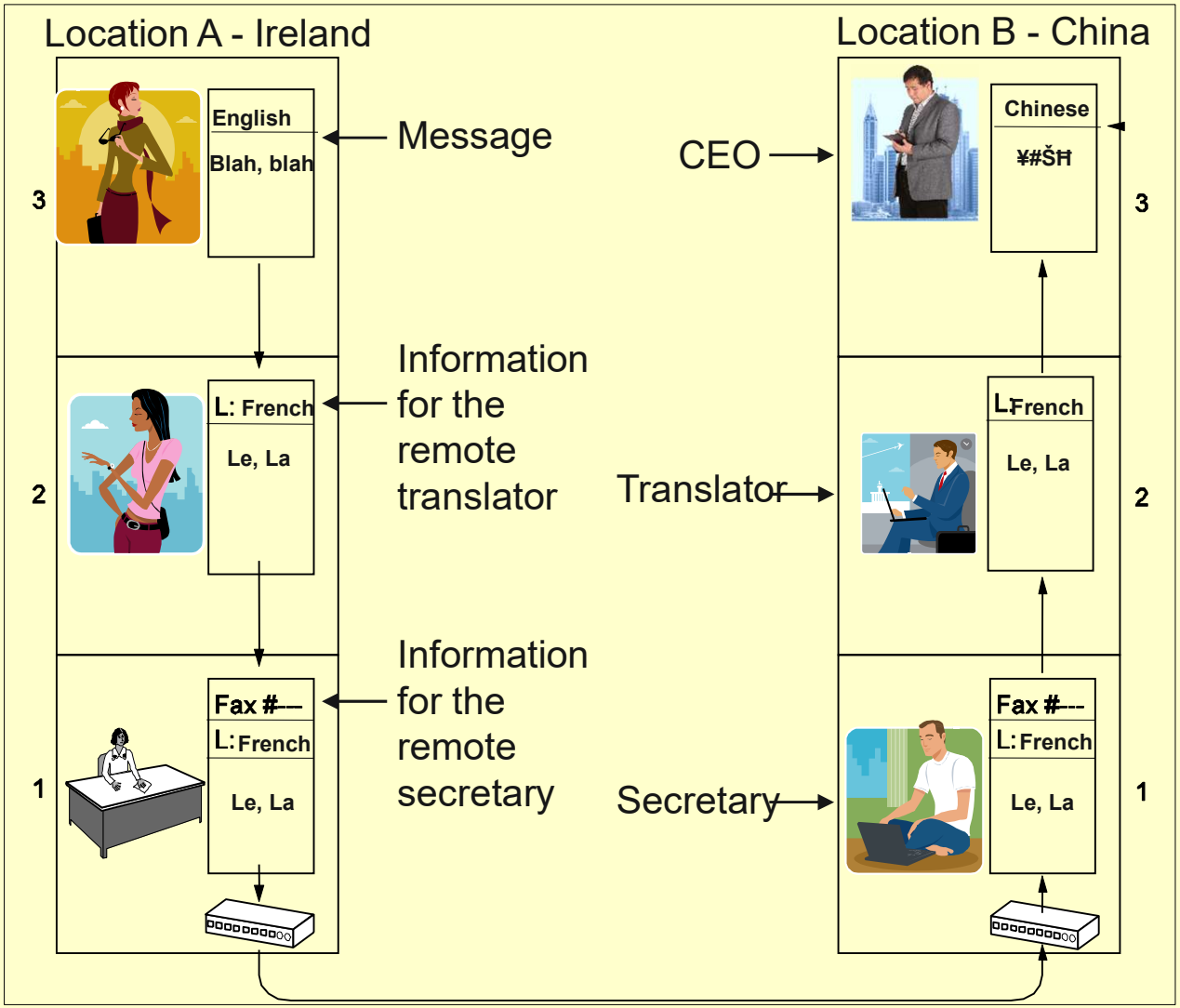
The *ISO OSI Reference Model*



Protocol Hierarchies - terminology

- Some terminology:
  - *Protocol/Protocol Entity.* A software module that performs one sub-task such as: *frame transmission/reception* etc.
  - *Protocol suites/stack*. A number of such software modules that work together to perform <u>all</u> of the tasks required to enable two hosts communicate.
  - *Protocol Architectures/Model*. A model by which we can categorize or view the different protocol layers.
- The following slide provides an analogy to the Communications Problem:
  - *How can two CEOs communicate with each other effectively and efficiently?*

The CEO-Translator-Secretary Architecture



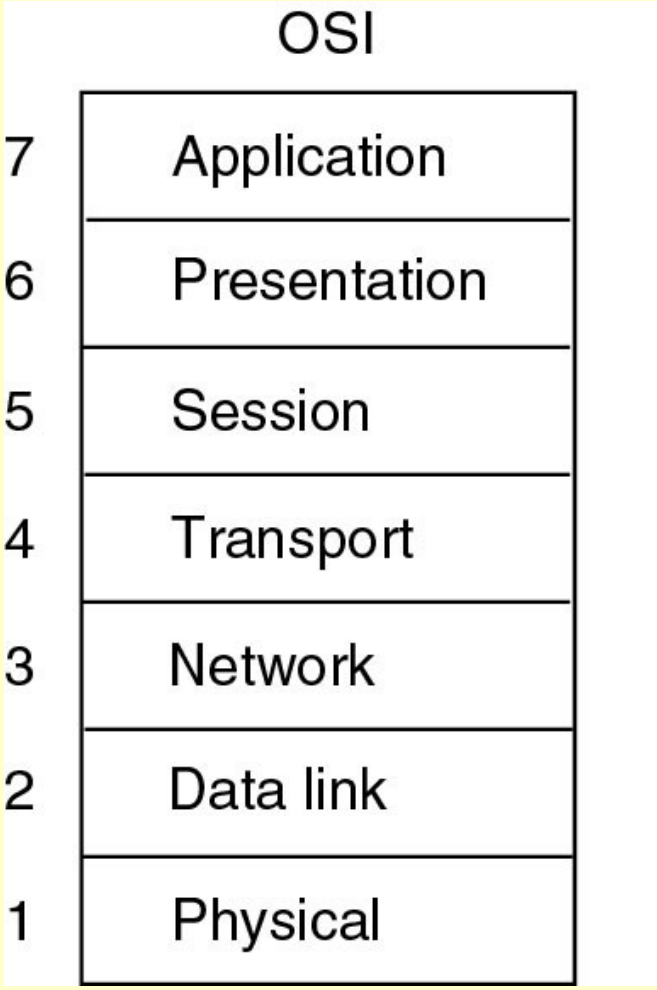
The OSI ISO Reference (7-Layer) Model

- Communications between computer hosts presents the same challenges.
- The *OSI model* deals with connecting open systems:
  - i.e. systems that are open for communication with other systems regardless of who built them or how they were manufactured.
- The principles that were applied to arrive at the seven layers are as follows:
  - Each layer was created when a different ***level of abstraction*** was required.
  - Each layer performs one well-defined function with each function chosen carefully to facilitate the development of ***standardized protocols***.
  - The number of layers chosen was sufficient enough to ensure that distinct functions were not *lumped* together to become unwieldy.

The OSI ISO Reference (7-Layer) Model

- The layers of software work together in unison to provide overall solutions to the problems of communicating between host computers across complex networks.
- ***Information*** flows across each of the interfaces (boundaries) between the layers:
  - These boundaries were carefully defined to *minimize* information flow across the interfaces.
  - The following slides discuss the functionality of each layer of protocol software. This functionality should be recognisable from previous discussions.

The *ISO OSI Reference Model*



The **7-Layer *Model: Layer-by-layer***

- The *Physical* Layer (Layer 1): Concerned with transmitting raw bits over a *communication channel*:
  - Its purpose is to ensure that the transmission of binary 1s and 0s adheres to what is appropriate for the transmission medium and what is expected by the *receiver.*
  - Primarily it deals with matters such as: What voltage levels are used? What frequencies are used? How long does it take to transmit a bit (Bit Duration)? etc.
  - There are certain design issues to address such as: the *mechanical and electrical* design of the plugs and sockets, *timing interfaces*, the physical *transmission medium* etc.

The **7-Layer *Model: Layer-by-layer***

- The *Data Link* Layer (Layer 2): Concerned with the successful transmission of data across an individual link:
  - Its purpose is to transforms a raw transmission facility into a *data communications channel* that *appears* free of transmission errors.
  - Primarily it deals with matters such as the transmission/reception of *data frames.*
  - There are certain design issues to address such as: the creation of localised unique addressing, the creation of a unique framing structure, flow control, error control, controlling access to a shared channel etc.

The **7-Layer *Model: Layer-by-layer***

- The *Network* Layer (Layer 3): Concerned with the successful delivery of data between hosts across complex network infrastructures:
  - Its purpose is to control the operation of the *sub-networks* to achieve host-to-host delivery.
  - Primarily it deals with matters such as the routing of *packets* from the *source* station towards the *destination* station across sub-nets etc.
  - There are certain design issues to address such as: the creation a globally unique address space, the creation of a globally unique framing structure.
  - Essentially this layer is responsible for managing data communications across interconnected *heterogeneous* networks.

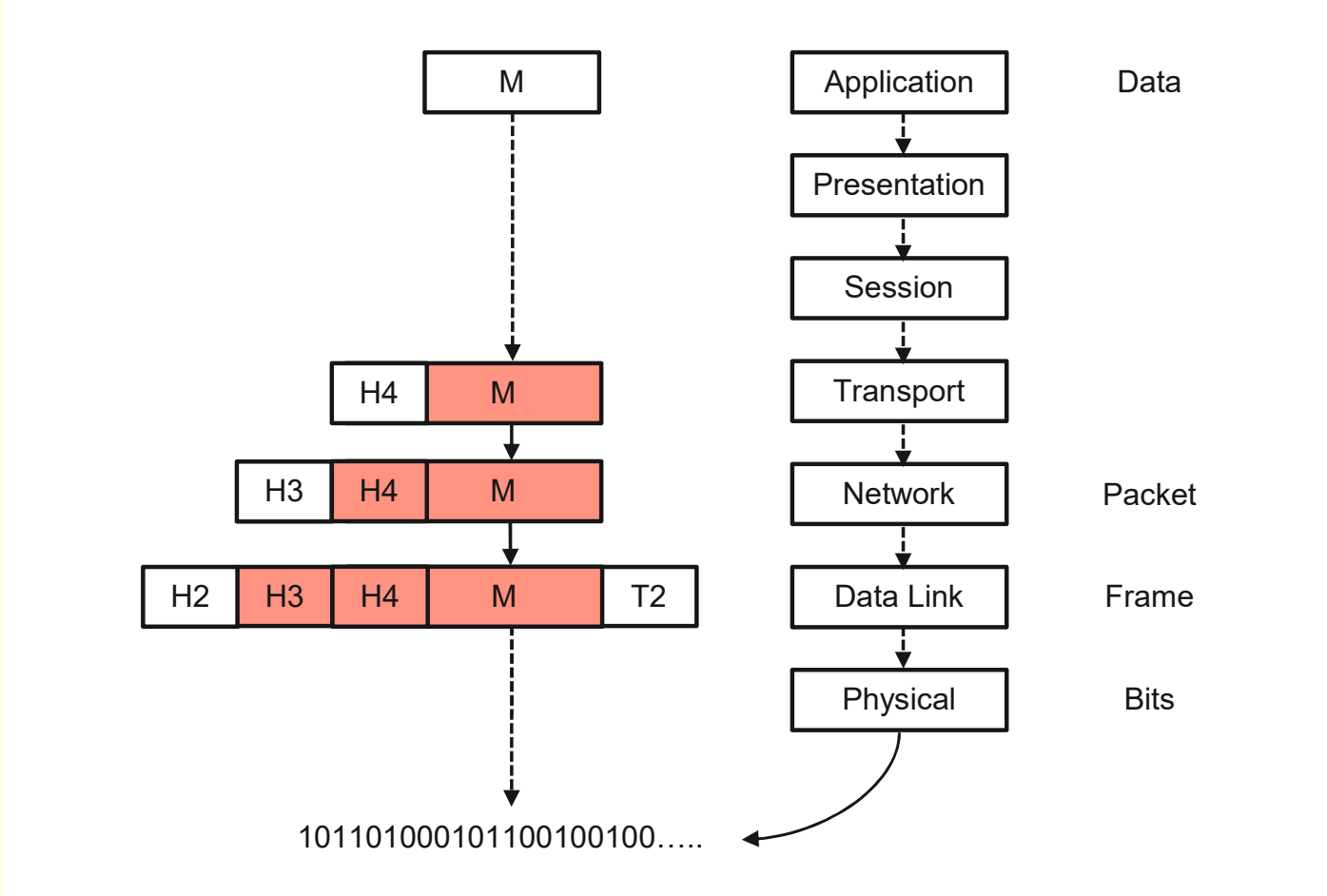
The **7-Layer *Model: Layer-by-layer***

- The *Transport* Layer (Layer 4): This is a key layer.
- It is a true *end-to-end* layer responsible for the reliable delivery of data between **processes** on end-host machines i.e. process-to-process delivery:
  - Its purpose is to provide a reliable data transport service to applications.
  - Primarily it deals with interfacing with applications for the purpose of exchanging data between applications across a network.
  - There are certain design issues to address such as: multiplexing data streams from/to remote applications, data loss, network latency etc.

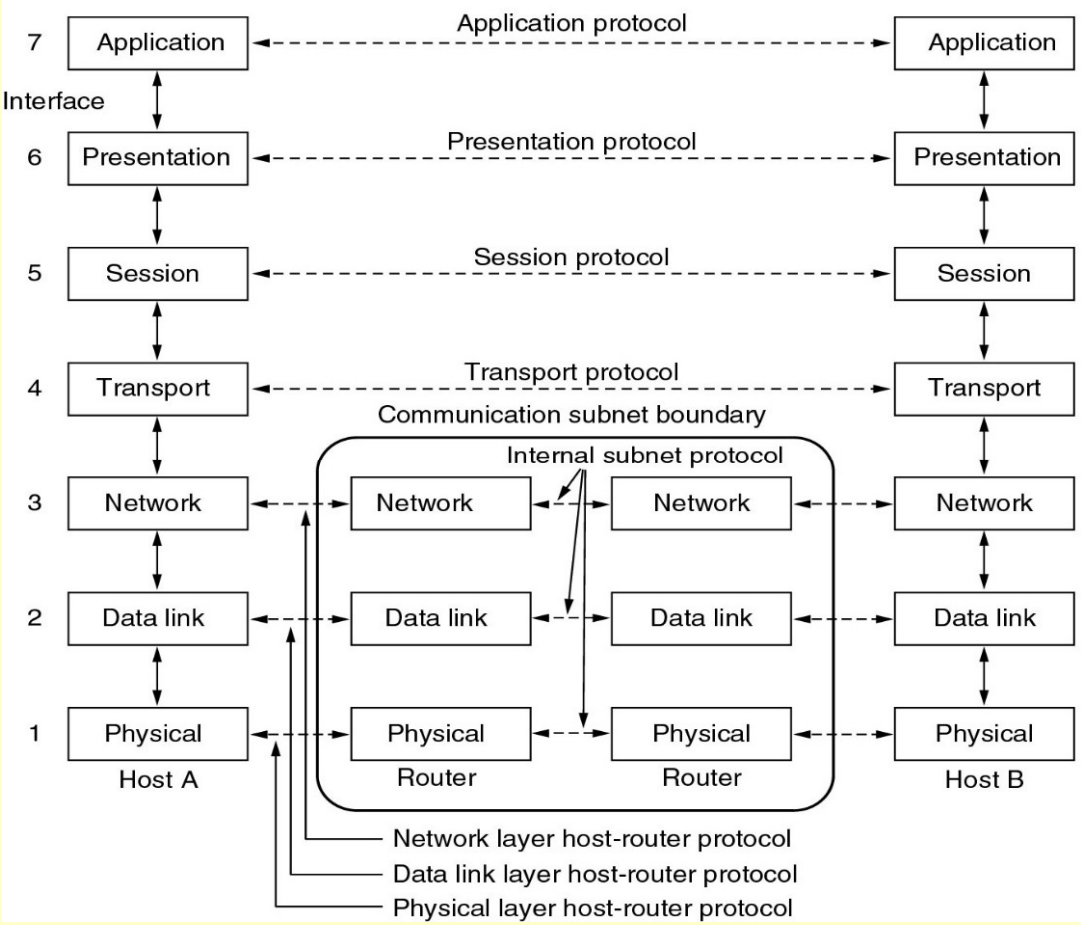
Data Flows

- The layers work together to provide the complete functionality required to facilitate communications between two remotely connected host machines regardless of how they connect to the internet.
- The following slide shows how data flows between the layers (up-and-down):
  - Note that sometimes it is necessary to break up the data to meet some size restrictions associated with some lower layer protocol software. We will discuss again when addressing the concept of *IP Fragmentation* at the Network layer.

Encapsulation and Information flow between the layers on an end-host.



Data Flows

- Some of the layers are also implemented on intermediary networking devices such as Routers:
  - This is because the functionality associated with these layers is needed for a particular purpose such as: frame reception/transmission, packet routing etc.
- The following slide shows how the complete set of layers are implemented on the end-hosts as well as on the intermediary networking devices.

The *ISO Reference Model* – Layers 1-3 V Layers 4-7